

MA1521 L1 Week 02 Wed 20260819

For what value(s) of c will $\frac{1}{5}\mathbf{i} - \frac{1}{5}\mathbf{j} + c\mathbf{k}$ be a unit vector?

A vector $\mathbf{A}$ has magnitude 6 and direction $(\mathbf{i} + 2\mathbf{j} - 2\mathbf{k})/3$.
If its tail is at $(-2, 0, 1)$, where is its head?

- a) Let P and Q be two points in space, and X the midpoint of the line segment PQ . Let O be an arbitrary fixed point; show that as vectors, $OX = \frac{1}{2}(OP + OQ)$.
- b) With the notation of part (a), assume that X divides the line segment PQ in the ratio $r : s$, where $r + s = 1$. Derive an expression for OX in terms of OP and OQ .

Find the angle between the vectors

- a) $\mathbf{i} - \mathbf{k}$ and $4\mathbf{i} + 4\mathbf{j} - 2\mathbf{k}$ b) $\mathbf{i} + \mathbf{j} + 2\mathbf{k}$ and $2\mathbf{i} - \mathbf{j} + \mathbf{k}$.

Tell for what values of c the vectors $c\mathbf{i} + 2\mathbf{j} - \mathbf{k}$ and $\mathbf{i} - \mathbf{j} + 2\mathbf{k}$ will

- a) be orthogonal b) form an acute angle

Calculate the value of the determinants a) $\begin{vmatrix} 1 & 4 \\ 2 & -1 \end{vmatrix}$ b) $\begin{vmatrix} 3 & -4 \\ -1 & -2 \end{vmatrix}$

Calculate $\begin{vmatrix} -1 & 0 & 4 \\ 1 & 2 & 2 \\ 3 & -2 & -1 \end{vmatrix}$ using the Laplace expansion by the cofactors of:

a) the first row b) the first column

Find $\mathbf{A} \times \mathbf{B}$ if

a) $\mathbf{A} = \mathbf{i} - 2\mathbf{j} + \mathbf{k}$, $\mathbf{B} = 2\mathbf{i} - \mathbf{j} - \mathbf{k}$ b) $\mathbf{A} = 2\mathbf{i} - 3\mathbf{k}$, $\mathbf{B} = \mathbf{i} + \mathbf{j} - \mathbf{k}$.

Find the area of the triangle in space having its vertices at the points

$$P : (2, 0, 1), \quad Q : (3, 1, 0), \quad R : (-1, 1, -1).$$

Find the equations of the following planes:

- a) through $(2, 0, -1)$ and perpendicular to $\mathbf{i} + 2\mathbf{j} - 2\mathbf{k}$
- b) through the origin, $(1, 1, 0)$, and $(2, -1, 3)$
- c) through $(1, 0, 1)$, $(2, -1, 2)$, $(-1, 3, 2)$
- d) through the points on the x , y and z -axes where $x = a$, $y = b$, $z = c$ respectively
- e) through $(1, 0, 1)$ and $(0, 1, 1)$ and parallel to $\mathbf{i} - \mathbf{j} + 2\mathbf{k}$

Find in parametric form the equations for

- a) the line through $(1, 0, -1)$ and parallel to $2\mathbf{i} - \mathbf{j} + 3\mathbf{k}$
- b) the line through $(2, -1, -1)$ and perpendicular to the plane $x - y + 2z = 3$
- c) all lines passing through $(1, 1, 1)$ and lying in the plane $x + 2y - z = 2$

Where does the line through $(0, 1, 2)$ and $(2, 0, 3)$ intersect the plane $x + 4y + z = 4$?

The line passing through $(1, 1, -1)$ and perpendicular to the plane $x + 2y - z = 3$ intersects the plane $2x - y + z = 1$ at what point?